

Homework 4

Incentive-Centered Design

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Solutions

1. (10+10+5+5=30 Points) Consider a unit-demand setting with $n = 2$ players and $k = 3$ goods. Denote by $v_i(j)$ the valuation of Player i for item j .
 - (a) Design a sealed bid DSIC auction that is individually rational and maximizes surplus (you have to design an allocation rule that maps bids to allocations and a payment rule that maps bids to payments).
 - (b) Suppose $v_1(1) = 1, v_1(2) = 7, v_1(3) = 8$, and Player 2's valuations are $v_2(1) = 2, v_2(2) = 3, v_2(3) = 5$. Determine all Walrasian equilibria for this instance. Depict the set of all WE prices in a graph.
 - (c) For the DSIC auction that you have designed in Part (a) and the instance specified in Part (b), compute the equilibrium allocation and equilibrium prices of goods. Is this a Walrasian equilibrium?
Remark: Here, by price of an item j , we mean the payment made by the bidder to whom item j is allocated by the sealed bid auction.
 - (d) A well-known iterative auction for unit-demand bidders is the Crawford-Knoer auction defined by the following pseudocode.

Algorithm 1 The Crawford-Knoer Auction for Unit-demand bidders

- 1: Initialize price of each item as $p(j) \leftarrow 0$ for all $j \in \{1, \dots, k\}$ and initialize all players as "Unassigned."
 - 2: **while** 1 **do**
 - 3: Ask every player to bid a favourite good (or ϕ) at prices $p(j) + \epsilon, j = 1, \dots, k$.
 - 4: If no bid is received from any unassigned player (which includes the case that there are no unassigned players), **HALT**.
 - 5: Else, choose an arbitrary unassigned bidder i that bid for item j and assign the item to i . If the item was previously assigned to any other player r , unassign r and increase price of j by ϵ .
 - 6: **end while**
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In the above pseudocode, favourite good of Player i at price $\mathbf{p}$ is defined as the good j^* such that for all $j \in \{1, \dots, k\}$, $v_i(j^*) - p(j^*) \geq v_i(j) - p(j)$. Favourite good need not be unique, and while in general players can break ties arbitrarily, assume for the purposes of this assignment that players break ties in favor of goods with lower index.

It can be shown that the Crawford-Knoer auction is Ex Post Incentive Compatible, i.e., sincere bidding by all players is an Ex Post Nash Equilibrium. Determine the equilibrium allocation and prices computed by the Crawford-Knoer auction at termination for the instance specified in Part (b). Is that a Walrasian equilibrium?

Solution:

(a) Designing a DSIC, IR, Welfare-Maximizing Auction

We use the VCG (Clarke Pivot) mechanism.

Allocation Rule:

Given bids $b_i(j)$, choose an allocation maximizing total reported welfare:

$$x^*(b) \in \arg \max_x \sum_{i=1}^2 \sum_{j=1}^3 b_i(j) x_{ij}$$

subject to:

$$x_{ij} \in \{0, 1\}$$

$$\sum_{j=1}^3 x_{ij} \leq 1 \quad \forall i$$

$$\sum_{i=1}^2 x_{ij} \leq 1 \quad \forall j$$

Thus, the allocation is the maximum-weight matching.

Payment Rule:

For player i ,

$$p_i(b) = \left(\max_{x^{-i}} \sum_{r \neq i} \sum_j b_r(j) x_{rj} \right) - \left(\sum_{r \neq i} \sum_j b_r(j) x_{rj}^* \right)$$

That is, player i pays the externality imposed on the other players.

Properties:

- VCG is dominant strategy incentive compatible (DSIC).
- It is individually rational (IR).
- Since the allocation maximizes total valuation, it maximizes social surplus.

(b) Determining All Walrasian Equilibria

The valuations are:

	1	2	3
v_1	1	7	8
v_2	2	3	5

Efficient Allocation

Compute total welfare for feasible allocations:

$$1 \rightarrow 2, 2 \rightarrow 3 : \quad 7 + 5 = 12$$

$$1 \rightarrow 3, 2 \rightarrow 2 : \quad 8 + 3 = 11$$

$$1 \rightarrow 3, 2 \rightarrow 1 : \quad 8 + 2 = 10$$

Hence the welfare-maximizing allocation is:

$$\boxed{1 \rightarrow 2, \quad 2 \rightarrow 3}$$

with item 1 unallocated.

Let prices be (p_1, p_2, p_3) .

In a Walrasian equilibrium, every unallocated item must have price 0. Since item 1 is unallocated,

$$p_1 = 0$$

Also, Walrasian equilibrium prices are nonnegative, so

$$p_2 \geq 0, \quad p_3 \geq 0.$$

Player 1's utility constraints

Player 1 receives item 2, so:

$$7 - p_2 \geq 1 - p_1$$

$$7 - p_2 \geq 8 - p_3$$

$$7 - p_2 \geq 0$$

Substituting $p_1 = 0$ gives:

$$p_2 \leq 6$$

$$p_3 - p_2 \geq 1$$

$$p_2 \leq 7$$

Player 2's utility constraints

Player 2 receives item 3, so:

$$5 - p_3 \geq 2 - p_1$$

$$5 - p_3 \geq 3 - p_2$$

$$5 - p_3 \geq 0$$

Substituting $p_1 = 0$ gives:

$$p_3 \leq 3$$

$$p_3 - p_2 \leq 2$$

$$p_3 \leq 5$$

Combining all inequalities:

$$\boxed{p_1 = 0, \quad 0 \leq p_2 \leq 2, \quad p_2 + 1 \leq p_3 \leq p_2 + 2}$$

Thus, all Walrasian equilibrium prices are given by the above region.

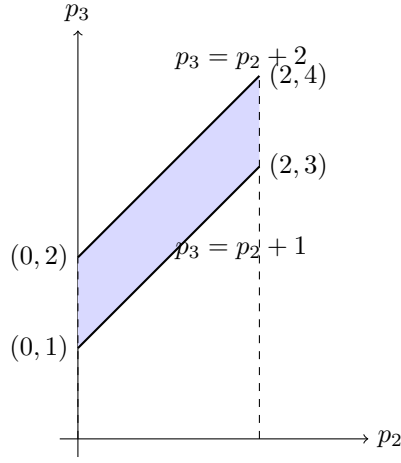
Examples include:

$$(0, 0, 1), \quad (0, 1, 2), \quad (0, 2, 3)$$

and all intermediate points satisfying the inequalities.

Hence the set of Walrasian equilibrium prices forms a convex region in the (p_2, p_3) plane.

The feasible region is depicted below.



Hence the set of Walrasian equilibrium prices forms a convex region in the (p_2, p_3) plane.

(c) Equilibrium Allocation and Prices in the DSIC Auction

The efficient allocation is:

$$1 \rightarrow 2, \quad 2 \rightarrow 3$$

VCG Payments

For Player 1:

Without Player 1, Player 2 gets item 3 with welfare:

$$W_{-1} = 5$$

With Player 1 present, Player 2 still gets welfare:

$$5$$

Hence:

$$p_1 = 5 - 5 = 0$$

For Player 2:

Without Player 2, Player 1 gets item 3 with welfare:

$$W_{-2} = 8$$

With Player 2 present, Player 1 receives item 2 with welfare:

$$7$$

Hence:

$$p_2 = 8 - 7 = 1$$

Following the convention in the problem statement, we interpret the payment made by the winner of an item as the price of that item.

Therefore the induced item prices are:

$$(p_1, p_2, p_3) = (0, 0, 1)$$

where item 2 is sold at price 0 and item 3 at price 1.

Since these prices satisfy the Walrasian equilibrium conditions derived in part (b), the VCG outcome is a Walrasian equilibrium.

Yes, this is a Walrasian equilibrium.

(d) Crawford–Knoer Auction

Initially:

$$(p_1, p_2, p_3) = (0, 0, 0)$$

and all players are unassigned.

At prices $p + \epsilon$:

$$u_1 = (1 - \epsilon, 7 - \epsilon, 8 - \epsilon)$$

so Player 1's favourite good is item 3.

Similarly,

$$u_2 = (2 - \epsilon, 3 - \epsilon, 5 - \epsilon)$$

so Player 2's favourite good is also item 3.

Suppose Player 1 is assigned item 3 first.

Then Player 2 bids for item 3, causing:

- Player 1 to become unassigned,
- the price of item 3 to increase by ϵ .

Each time Player 2 bids for item 3 while it is assigned to Player 1, Player 1 becomes unassigned and the price of item 3 increases by ϵ .

Player 1 continues preferring item 3 while

$$8 - p_3 > 7 - p_2$$

Since $p_2 = 0$ throughout the auction, this becomes

$$8 - p_3 > 7$$

that is,

$$p_3 < 1$$

When

$$p_3 = 1$$

Player 1 becomes indifferent between items 2 and 3:

$$7 - 0 = 8 - 1 = 7$$

By the tie-breaking rule favoring smaller indices, Player 1 chooses item 2.

At this price, Player 2 still strictly prefers item 3 since

$$5 - 1 = 4 > 3 - 0 = 3.$$

Hence the auction terminates with allocation:

$$1 \rightarrow 2, \quad 2 \rightarrow 3$$

and prices:

$$(p_1, p_2, p_3) = (0, 0, 1)$$

At prices $(0, 0, 1)$, Player 1 prefers item 2 while Player 2 prefers item 3, so no further reassignment occurs.

These prices satisfy the Walrasian equilibrium conditions from part (b).

Hence,

The Crawford–Knoer auction terminates at a Walrasian equilibrium.

2. (15 Points) Consider n unit-demand bidders and k items. Let $(M_1, \mathbf{q}_1)$ and $(M_2, \mathbf{q}_2)$ be two Walrasian Equilibria, where M_1 and M_2 are the “matchings” or allocations, and $\mathbf{q}_1$ and $\mathbf{q}_2$ are price vectors. Prove that $(M_1, \mathbf{q}_2)$ is also a Walrasian Equilibrium.

Solution:

Let $(M_1, \mathbf{q}_1)$ and $(M_2, \mathbf{q}_2)$ be two Walrasian equilibria in a unit-demand market.

We will prove that $(M_1, \mathbf{q}_2)$ is also a Walrasian equilibrium.

Recall that in a Walrasian equilibrium:

- every bidder receives a utility-maximizing item at the given prices,
- every unallocated item has price 0.

Since both $(M_1, \mathbf{q}_1)$ and $(M_2, \mathbf{q}_2)$ are Walrasian equilibria, both matchings M_1 and M_2 maximize total welfare.

Consider the symmetric difference graph

$$M_1 \oplus M_2.$$

Since each bidder and each item has degree at most 2, every connected component of this graph is either:

- an alternating cycle, or
- an alternating path.

We analyze these separately.

Case 1: Alternating cycle

Suppose the cycle is

$$i_1, j_1, i_2, j_2, \dots, i_t, j_t, i_1$$

where:

- bidder i_r receives item j_r in M_1 ,
- bidder i_r receives item j_{r-1} in M_2 .

Since $(M_2, \mathbf{q}_2)$ is a Walrasian equilibrium, bidder i_r weakly prefers j_{r-1} over j_r at prices $\mathbf{q}_2$:

$$v_{i_r}(j_{r-1}) - q_2(j_{r-1}) \geq v_{i_r}(j_r) - q_2(j_r).$$

Summing these inequalities over all bidders in the cycle gives

$$\sum_{r=1}^t v_{i_r}(j_{r-1}) - \sum_{r=1}^t q_2(j_{r-1}) \geq \sum_{r=1}^t v_{i_r}(j_r) - \sum_{r=1}^t q_2(j_r).$$

The same set of items appears on both sides, so the price terms cancel:

$$\sum_{r=1}^t v_{i_r}(j_{r-1}) \geq \sum_{r=1}^t v_{i_r}(j_r).$$

Since both M_1 and M_2 are welfare-maximizing allocations, the total welfare contributed by this cycle must be equal under both matchings. Hence equality must hold above:

$$\sum_{r=1}^t v_{i_r}(j_{r-1}) = \sum_{r=1}^t v_{i_r}(j_r).$$

Since each individual inequality is weak and their sum is equality, every inequality must itself hold with equality. Therefore for every bidder i_r ,

$$v_{i_r}(j_{r-1}) - q_2(j_{r-1}) = v_{i_r}(j_r) - q_2(j_r).$$

Thus bidder i_r is indifferent between the item assigned in M_2 and the item assigned in M_1 at prices $\mathbf{q}_2$.

Case 2: Alternating path

Any alternating path begins and ends at unmatched items.

Suppose the path is

$$j_0, i_1, j_1, i_2, \dots, i_t, j_t$$

where j_0 and j_t are unmatched endpoint items.

Since unmatched items in a Walrasian equilibrium have price 0, we have

$$q_2(j_0) = q_2(j_t) = 0.$$

Applying the same argument as in the cycle case and summing the utility inequalities along the path, the price terms telescope and cancel because the endpoint prices are 0.

Hence the total welfare under M_1 and M_2 along the path must again be equal. Therefore all inequalities along the path hold with equality.

Thus every bidder on the path is indifferent between their assignment in M_1 and in M_2 at prices $\mathbf{q}_2$.

Combining the two cases, every bidder weakly prefers their allocation in M_1 at prices $\mathbf{q}_2$. Hence every bidder receives a utility-maximizing item under prices $\mathbf{q}_2$ in allocation M_1 .

Furthermore, any item unmatched in M_1 appears as an endpoint of an alternating path in $M_1 \oplus M_2$. Since endpoint unmatched items have price 0 under $\mathbf{q}_2$, every item unmatched in M_1 also has price 0 under $\mathbf{q}_2$.

Hence all Walrasian equilibrium conditions are satisfied for $(M_1, \mathbf{q}_2)$.

Therefore,

$(M_1, \mathbf{q}_2)$ is also a Walrasian equilibrium.
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3. (Optional, 0 Points) Prove the theorem stated in Lecture 25, Slide 13 of the lecture slides.

Solution: We prove that the ascending auction for Scenario 1 is an Awesome Ascending Auction.

Recall that an ascending auction is Awesome if:

- it is Ex Post Incentive Compatible (EPIC), and
- it computes a welfare-maximizing allocation.

Part 1: EPIC

We show that sincere bidding is an Ex Post Nash Equilibrium.

Fix any bidder i and suppose all other bidders bid sincerely.

Recall that sincere bidding means:

$$\text{Bid "Yes" for item } j \iff v_{ij} \geq p(j).$$

Suppose bidder i deviates from sincere bidding.

There are two possible kinds of deviations.

Case 1: Bid "No" when $v_{ij} \geq p(j)$.

Then bidder i refuses an item that yields nonnegative utility:

$$v_{ij} - p(j) \geq 0.$$

If bidder i would eventually have obtained the item under sincere bidding, then refusing to bid may cause the bidder to lose a nonnegative-utility item.

Hence this deviation can only weakly decrease utility.

Case 2: Bid "Yes" when $v_{ij} < p(j)$.

If bidder i eventually wins item j at price $p(j)$, then the resulting utility is

$$v_{ij} - p(j) < 0.$$

Thus the bidder obtains negative utility.

If the bidder does not win the item, then the deviation has no effect on utility.

Hence this deviation cannot strictly improve utility.

Therefore no unilateral deviation can improve utility once all other bidders bid sincerely.

Hence sincere bidding constitutes an Ex Post Nash Equilibrium.

Therefore the auction is EPIC.

Part 2: Welfare Maximization

Fix an item j .

The auction repeatedly increases the price of item j while multiple bidders are willing to purchase it.

Since bidders bid sincerely, bidder i remains interested in item j precisely while

$$v_{ij} \geq p(j).$$

Hence bidder i drops out exactly when the price exceeds v_{ij} .

Therefore the final bidder remaining for item j is the bidder with the highest valuation for that item.

Moreover, the item is sold at a price equal to the second-highest valuation for that item.

Thus each item is allocated to the bidder who values it the most.

Since valuations are additive across items in Scenario 1, maximizing welfare is equivalent to independently assigning each item to the bidder with the highest valuation.

Therefore the resulting allocation maximizes total social welfare.

Hence the auction is:

- EPIC, and
- welfare-maximizing.

Therefore the auction is an Awesome Ascending Auction.

The auction is Awesome.